



see first
understand first
act first

EXPANDING SOA Through Enterprise Data Mashups



Enterprise Data
Mashups

Demonstrating the Real Value of SOA



CONSORTIUM™

Creating a Portfolio of Composite Reusable Data Services for SOA

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**SVP, North America
Denodo Technologies**

December 10-11, 2008

Agenda

- Enterprise Data Mashups – What & Why?
- Role of Mashups in Data Services and SOA
- Case Studies
 - Telco: Data Services for Enterprise SOA
 - Data Services for Web / WOA
 - Data Services for BI, Reporting, Compliance
- Enterprise Data Mashups
 - Technical Architecture
 - Co-existence with SOA Middleware
- Conclusions



Time To Information



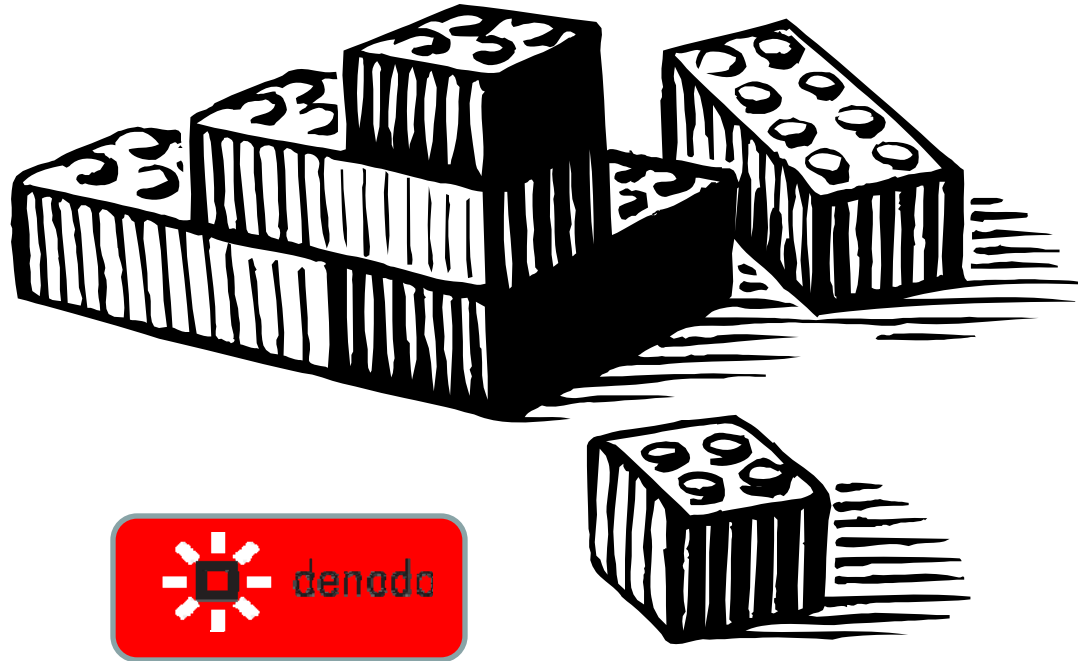
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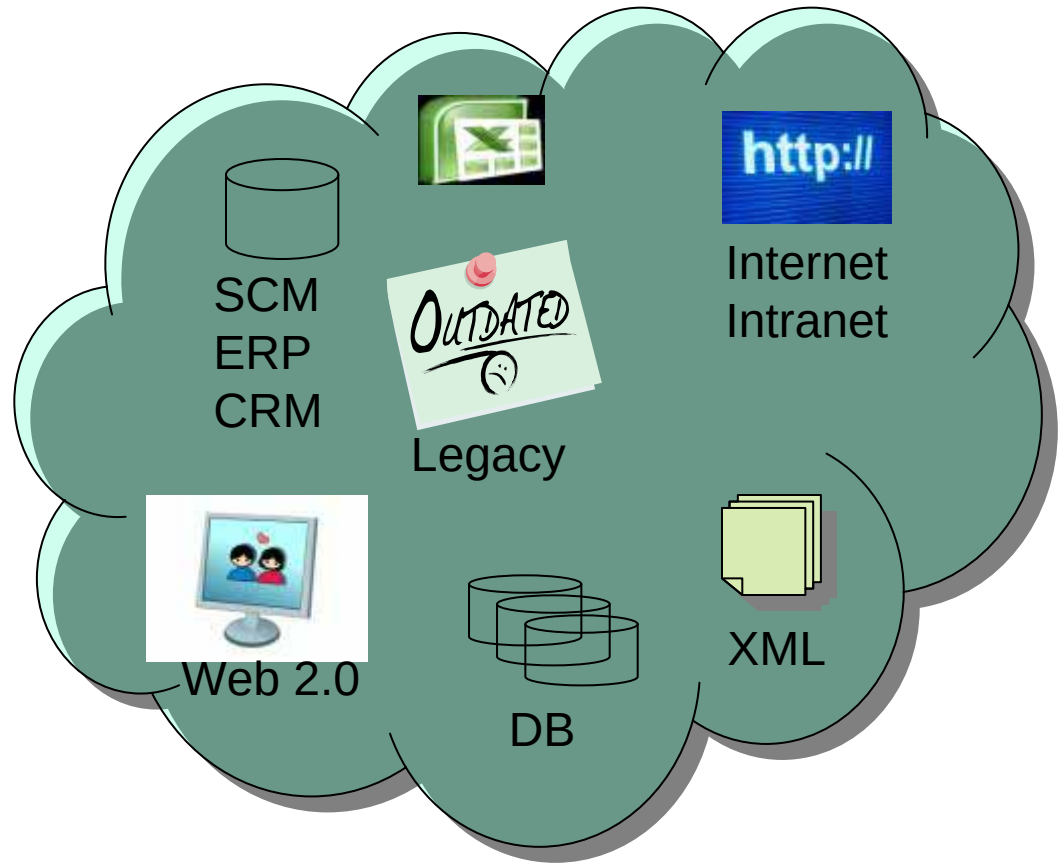


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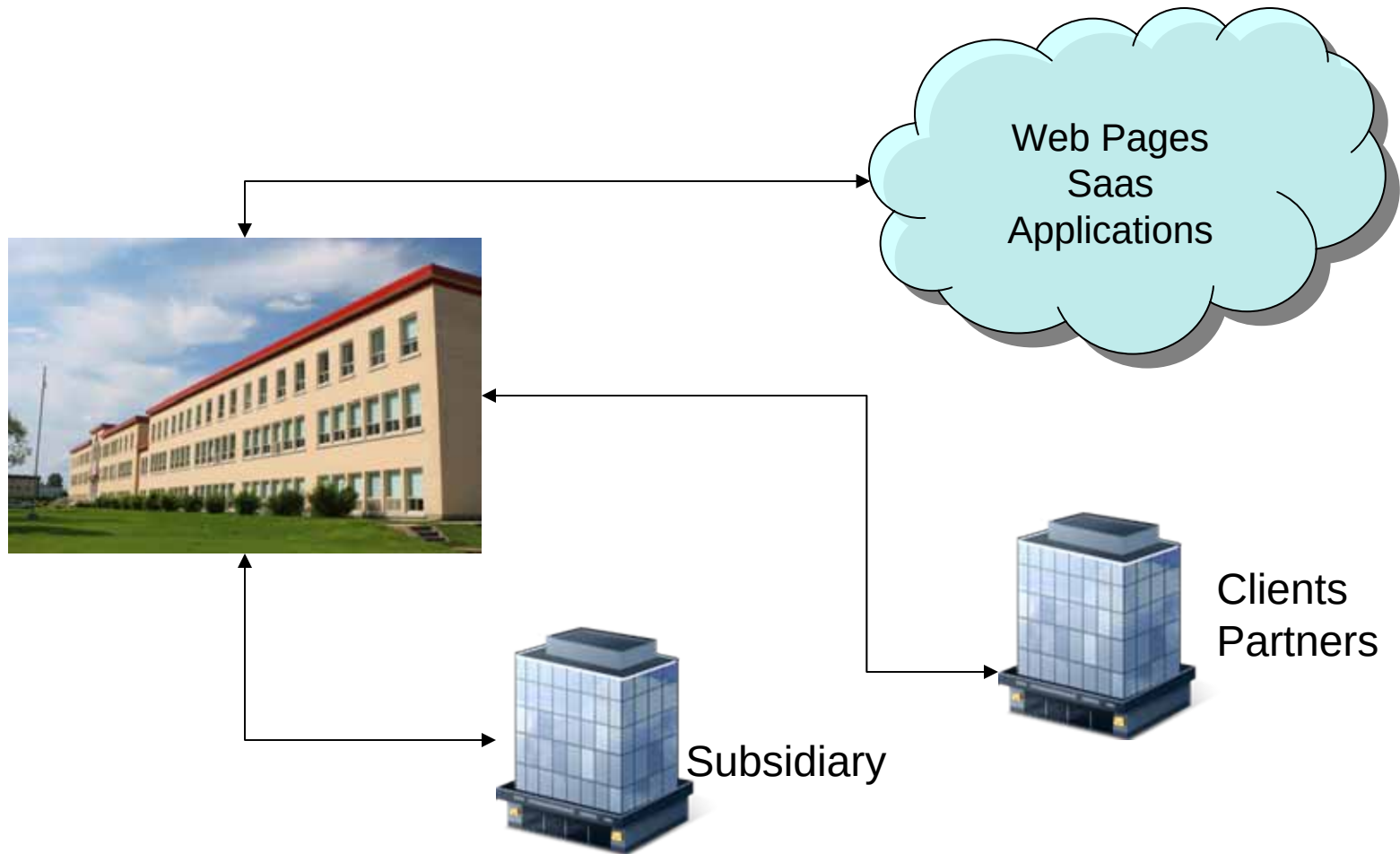
From Silo to Lego



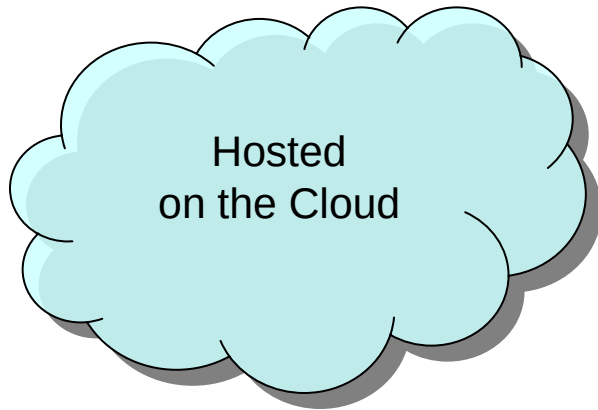
Explosion of Sources of Data



Look Where My Data Went

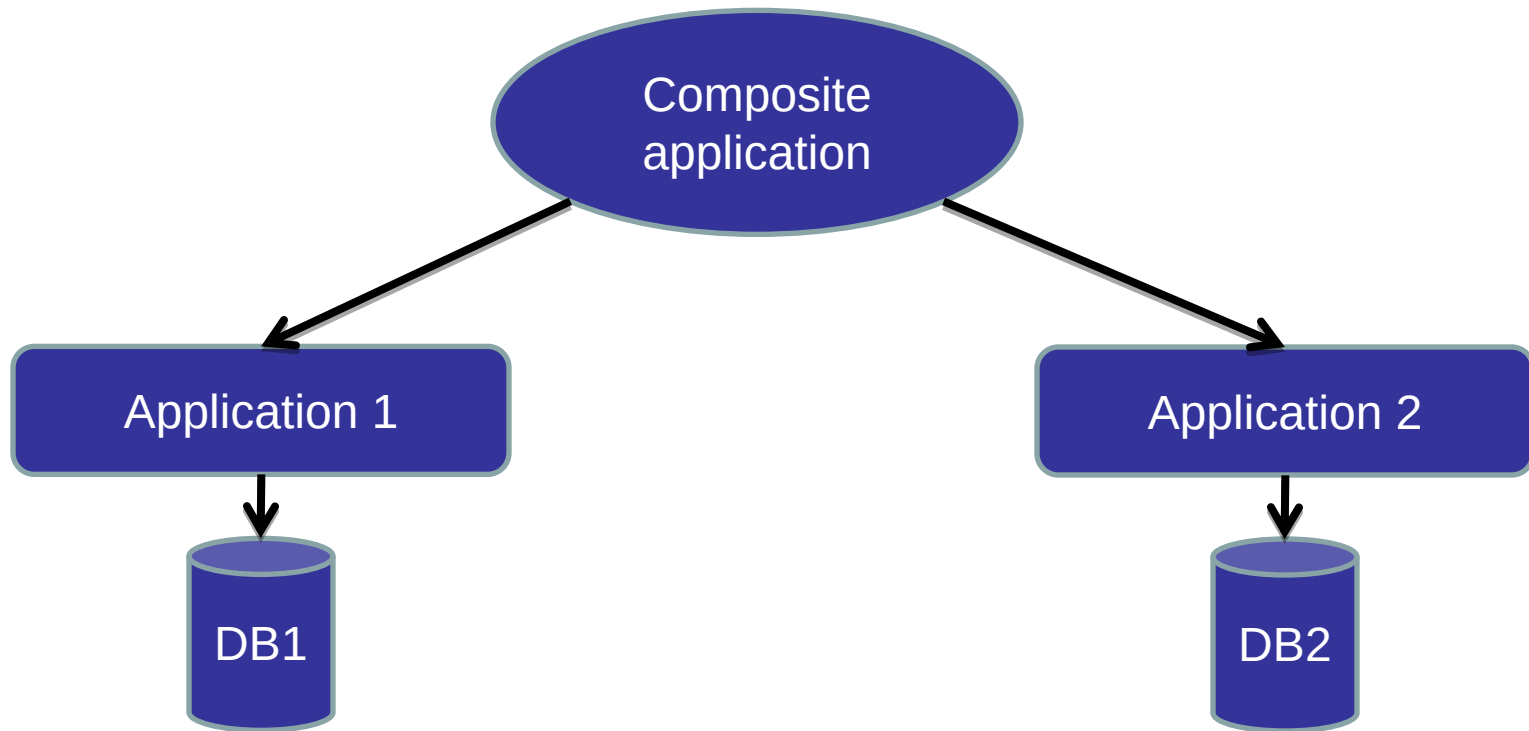


Tomorrow

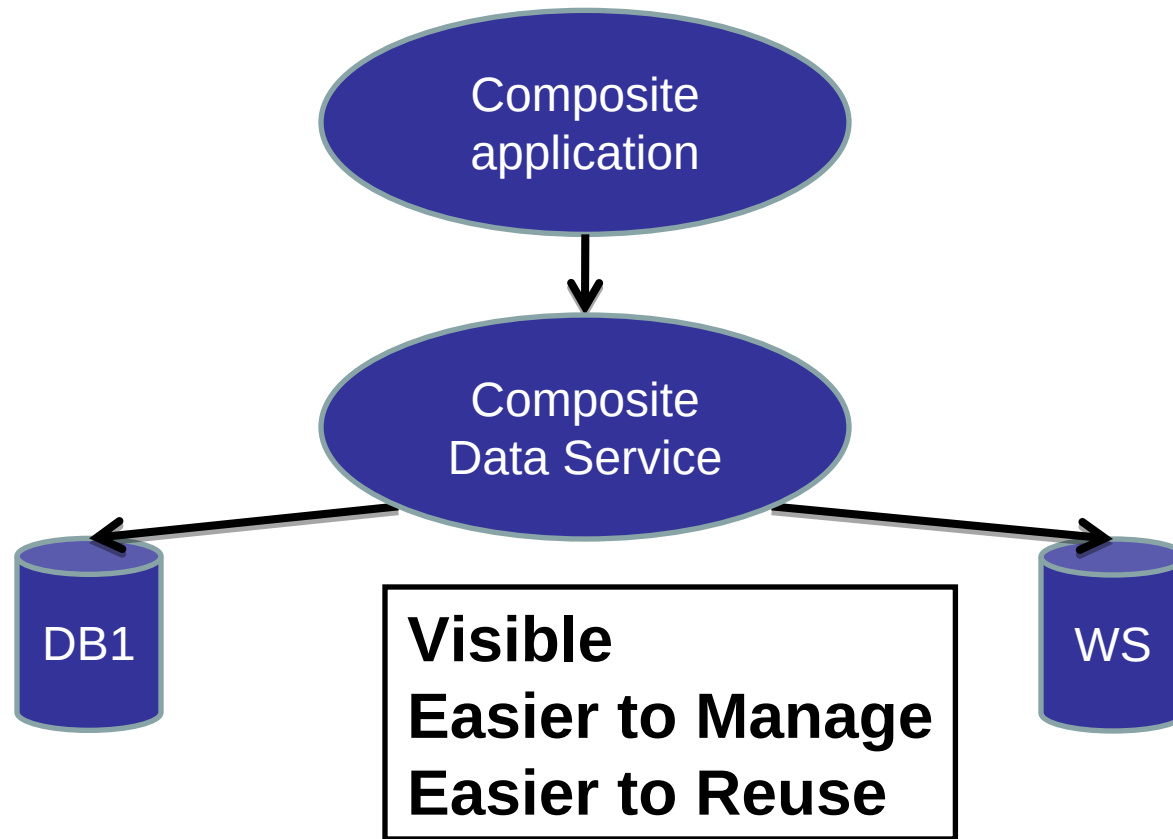


Shared Data Infrastructure

Can you Reuse a Composite Application?



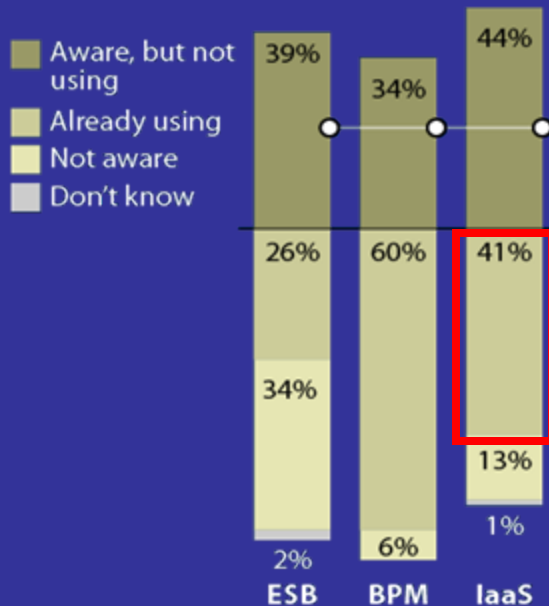
But Business knows how to reuse Composite Data Services



Forrester Research: Information as a Service (IaaS) / Data Service adoption is growing rapidly

SOA technology awareness

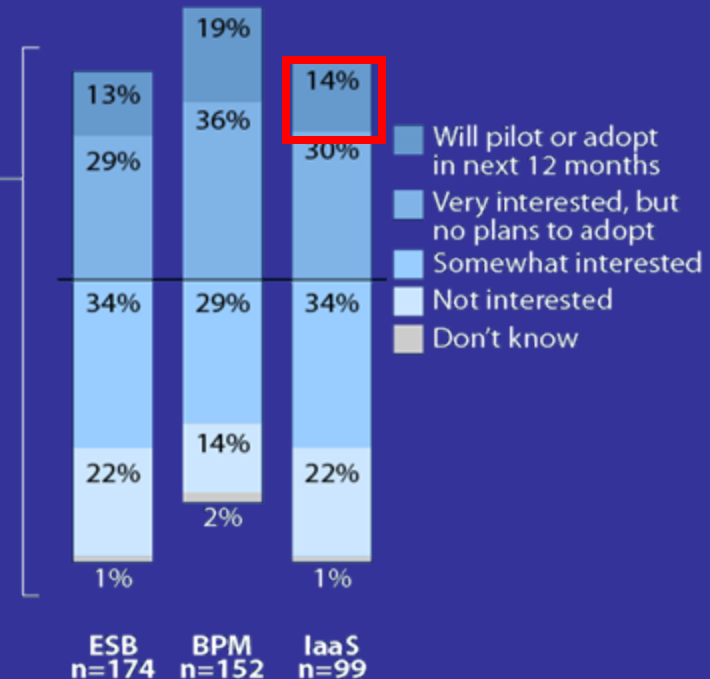
"How familiar are you with each of the following technologies and processes?"



Base: 449 decision-makers at North American and European enterprises who said adoption of SOA was either a priority or critical priority

SOA technology plans

"How interested are you in adopting each of the following technologies?"



Base: decision-makers at North American and European enterprises who were aware of the given technology, but not yet using it and SOA adoption was either a priority or critical priority

Source: Enterprise And SMB Software Survey, North America And Europe, Q3 2007



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Gartner: "SOA Isn't Just Transportation"

SOA Isn't Just Transportation in a Project or Domain

- Information sharing:
 - Identifying, filtering, alteration
- Internet and WANs:
 - Distinguishing between popularity and ease vs. reliability and validity
- Documents and content:
 - Separating fact, interpretation and context
- Applications:
 - Separating business process from data processing



True information sharing requires more formal process management for requirements, analysis, design and Implementation.

Gartner.



**Enterprise Data
Mashups**

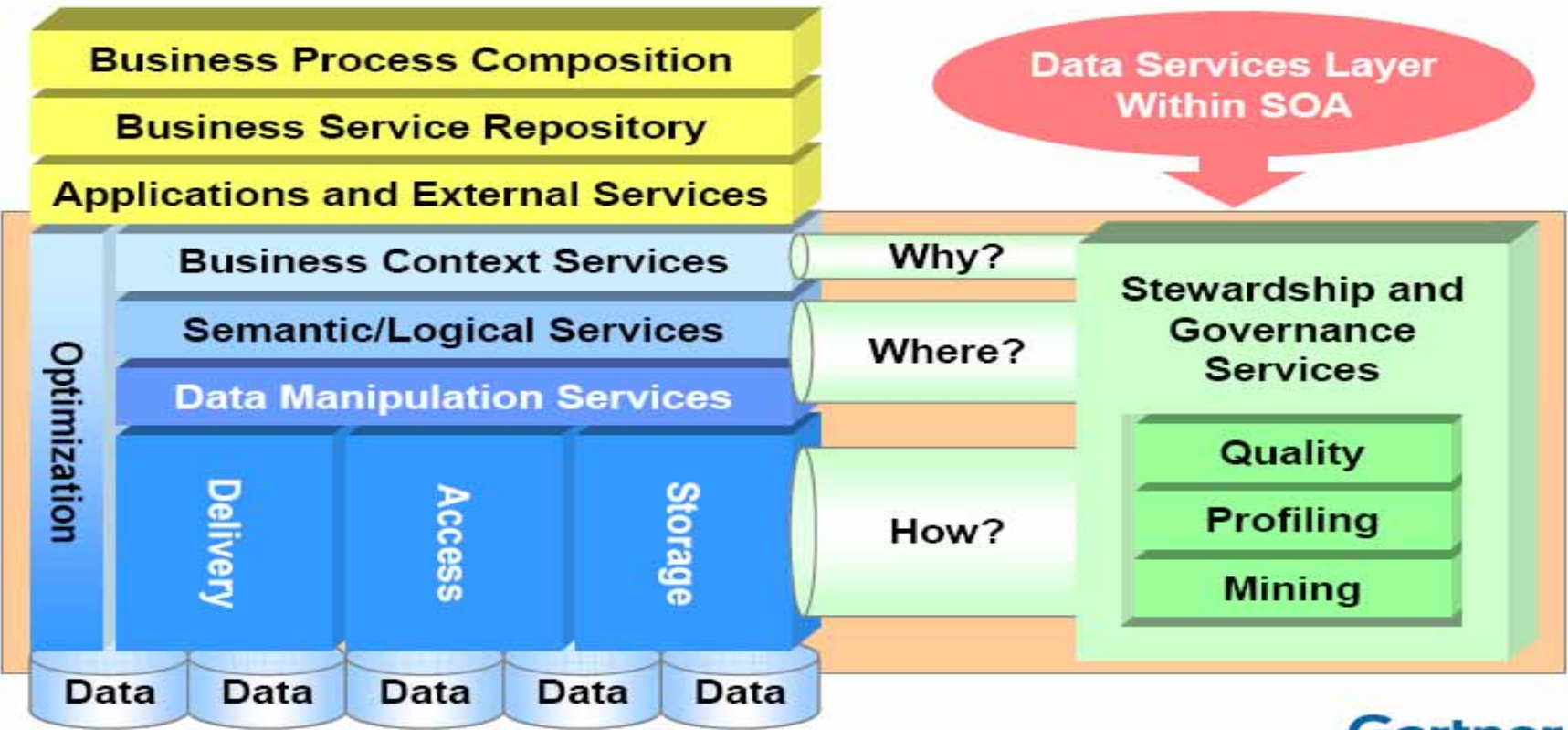
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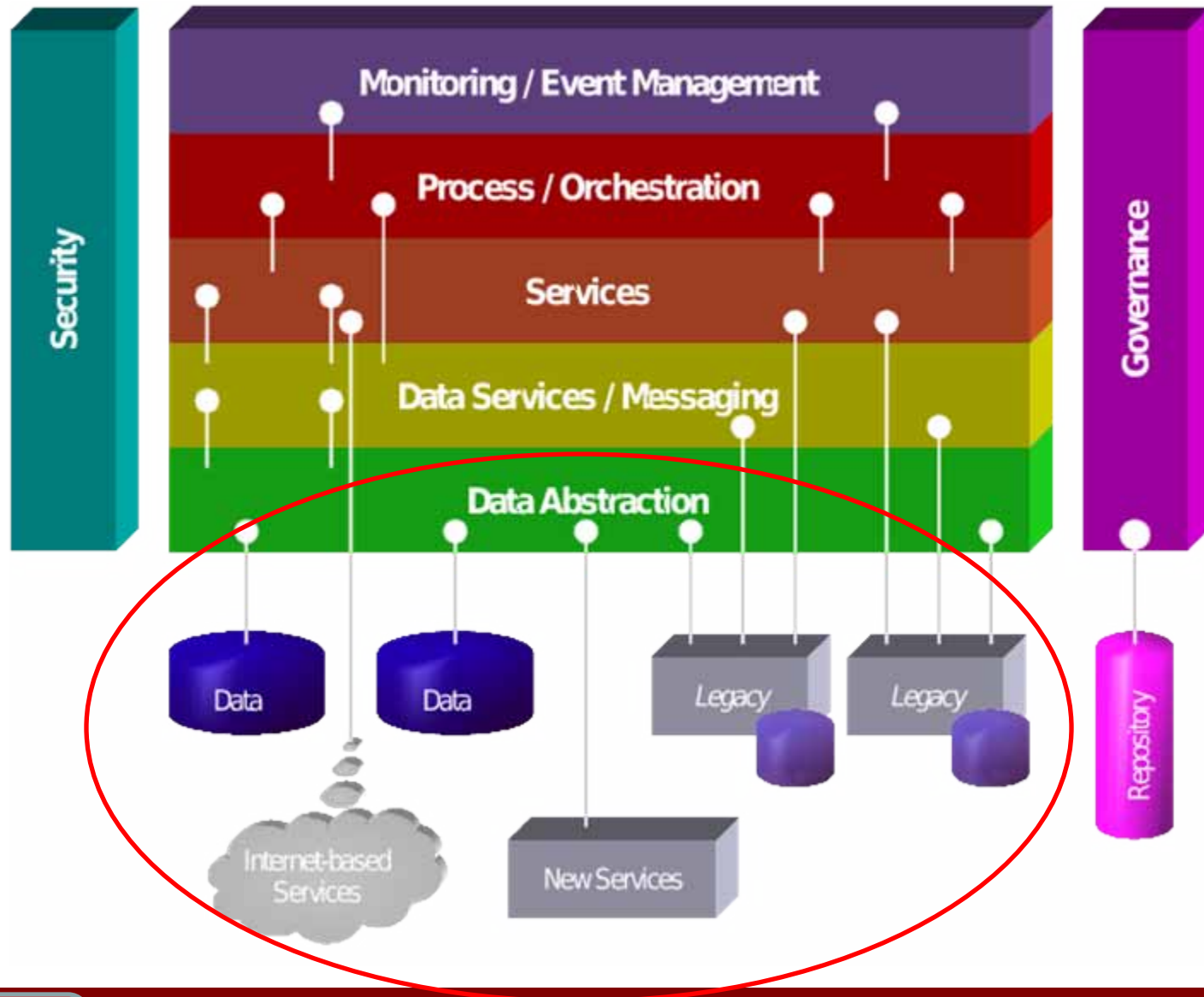
Gartner's View on Data Services

Data Services: The Intersection of Data Integration and SOA

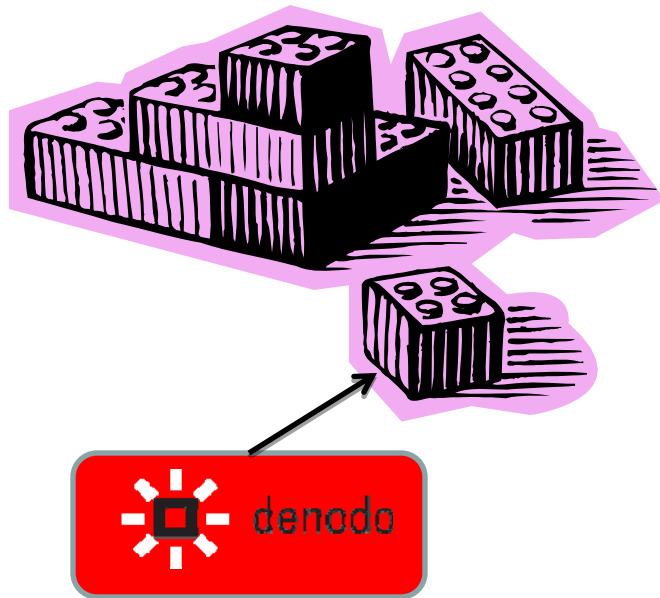


Gartner

Another View: Data Services & SOA Architecture



Enterprise Data Mashup



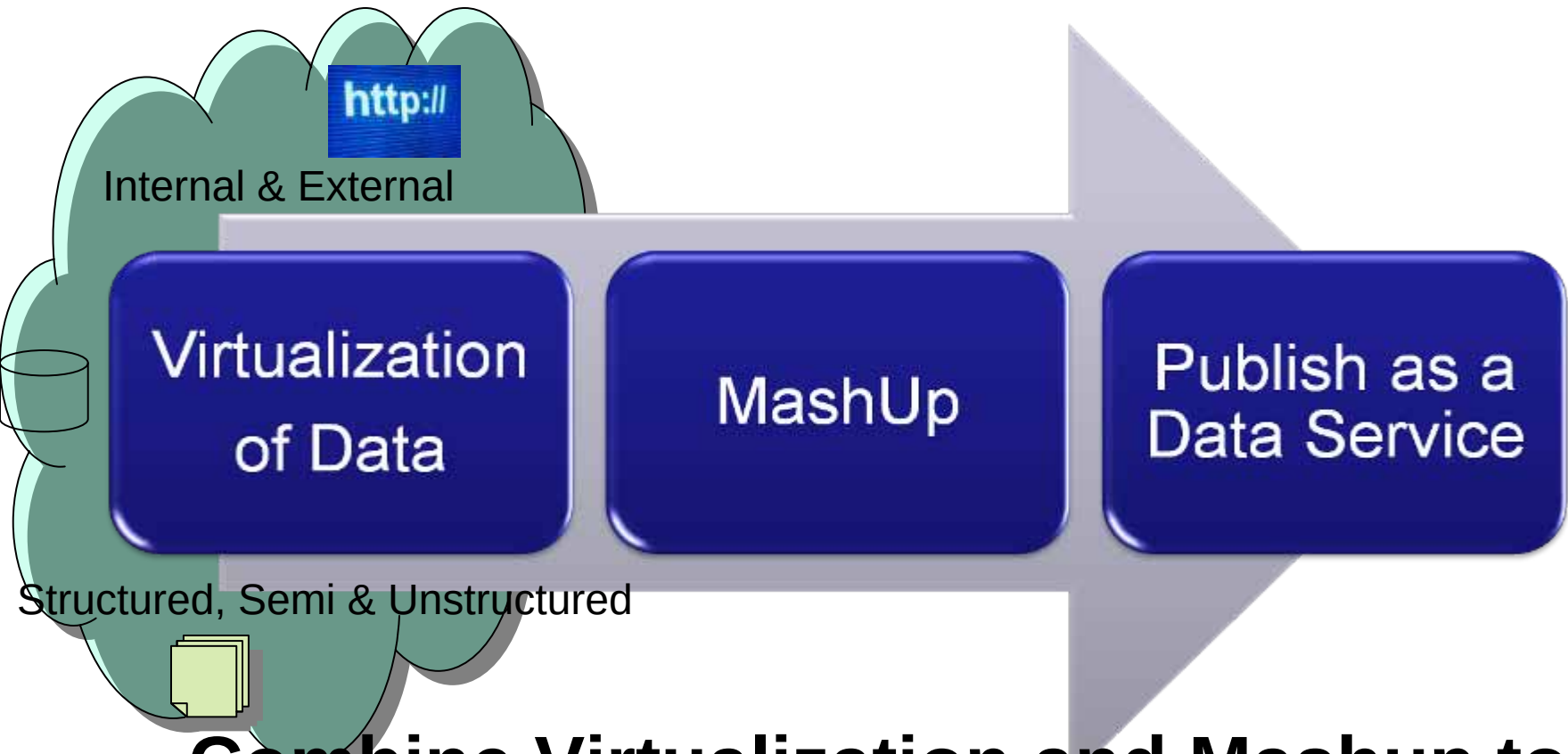
Data exposed as a Service

- Managed
- Registered
- Customizable

Fast and Fully Reusable

- No lines of code
- Deployed within a week
- Modified at will

Mashups Deliver Data Services



Combine Virtualization and Mashup to Abstract the Data into Services



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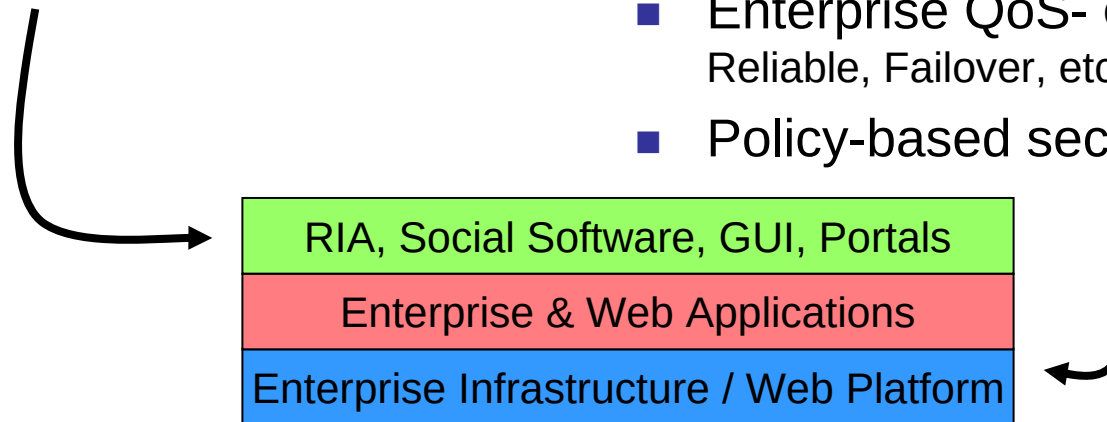
Web Mashups vs. Enterprise Data Mashups

Web Mashups

- Visual centric
- Data relationships are simple
- End-user driven enabled by simple APIs / tools
- No QoS guarantees – as is
- Simple Web security

Enterprise Data Mashups

- Data & Information centric
- Deep transformations & semantic relationships
- Interact with Enterprise ecosystem
- Business analyst driven w/ user control options
- Enterprise QoS- Query Optimization, Reliable, Failover, etc.
- Policy-based security



Denodo Enterprise Data Mashup Case Studies



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Demonstrating the Real Value of SOA



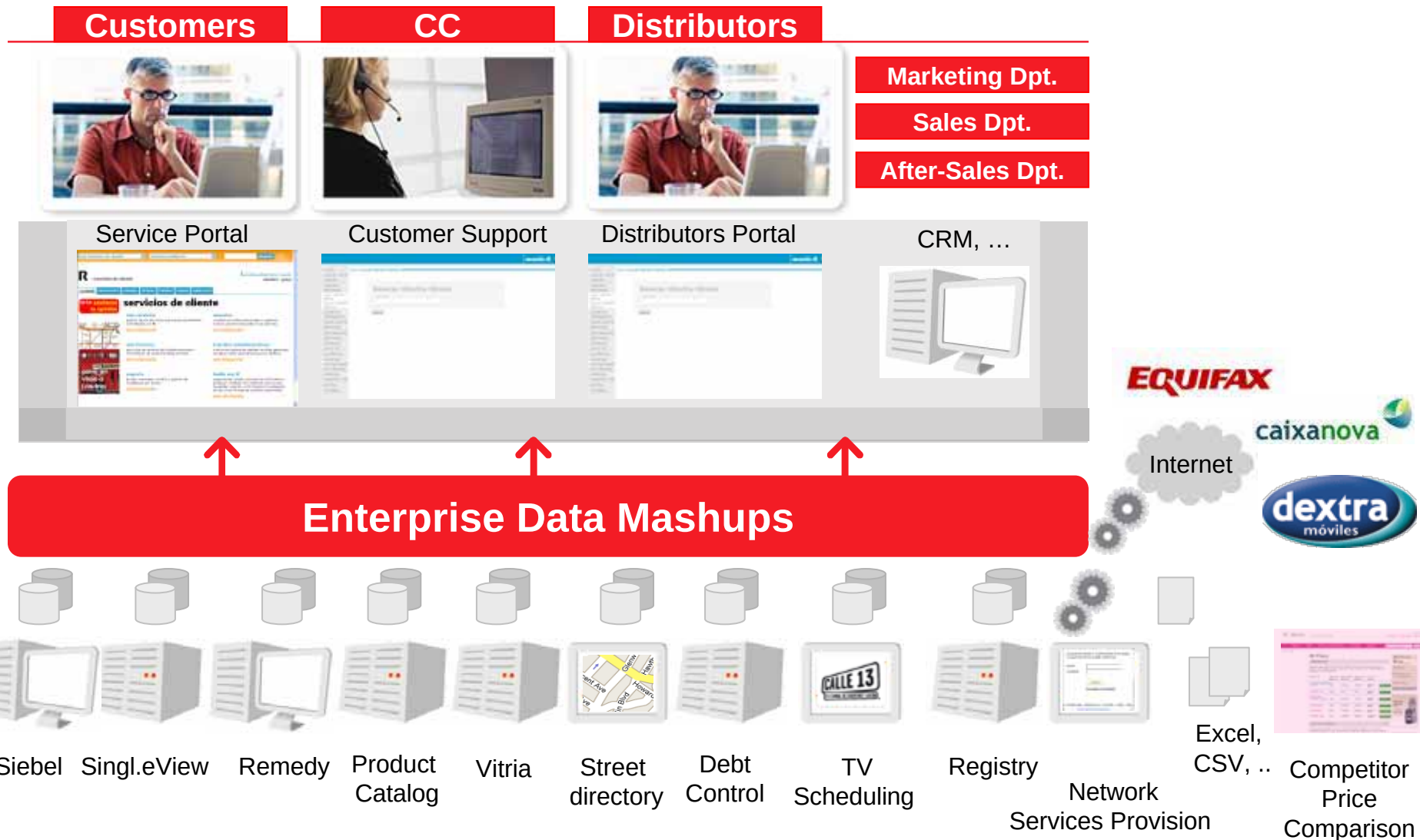
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Mashup Data Services – Denodo Case Study

Customer Self Service Portal for Cable Leader

- Regional Telecommunications Operator
 - Offers **quadruple-play** : Phone, TV, Broadband Internet & VMO
- More than 50% market share in territory.
 - **Competitive advantage**: Customer service & Product innovation
- Problem: Did not have the right infrastructure to Innovate:
 - **Complex architecture where the integration of different systems becomes a difficult and tedious task.**
 - Tried patch solutions for Web Mashups, but not scalable
- Result:
 - **Roadblocks and Delays** to launch new products/services.
- Need: **Agile creation of new data and mashup services**
 - For **internal** and **external consumption**.
 - **Reusing the existing IT infrastructure.**
 - Easy to **maintain** and **adapt** → highly changing environment.

Holistic View of Customer - Solution Architecture



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Information Sources (I)

- **Internal**

- **CRM** (Siebel): contracts, activities, incidences reported by customers, etc.
 - Database datasource.
- **Invoicing System** (Singl.eView):
 - Database datasource.
- **Service Management** (Remedy): incidences.
 - Database datasource.
- **Street Directory**:
 - Database datasource.
- **EAI** (Vitria): services provision, etc.
 - Database datasource.
- **Product Catalog**: products, services, equipments, etc.
 - Database datasource.
- **TV Scheduling**: channels, thematic, etc.
 - Database datasource.

Information Sources (II)

- **Internal**

- **Network Services Provision**: SIMs blocking/unblocking, etc.
 - Web interface.
- **Registry**: calling details, SMS, PPV events, etc.
 - Database datasource.
- **Debt control**: bills, no payments, etc.
 - Database datasource.
- **Files**: Excel, CSV, etc.

- **External**

- **www.equifax.com**: control defaulter clients .
 - Web interface.
- **On-line transactions** (Caixanova): bank.
 - Web interface.
- **Logistics Operator** (Dextra Moviles): mobile stock checking, orders checking, etc.
 - Web Services interface.

Data combination needs

- ✱ Combination of the products hired by the client with the detailed information of these products.
 - ✱ **Invoicing System + Product Catalog.**
- ✱ Combination of incidences information with their solving status.
 - ✱ **CRM + Service Management.**
- ✱ Combination of customer feedback to new products / plans.
 - ✱ **CRM Notes field & Email + Product Catalog.**
- ✱ Combination of internal and competitor pricing and offers.
 - ✱ **Competitor Web extraction + Product Catalog & Offers**

Comparison Views of Competitive Pricing & Terms

Detailed Product comparison

Vendor Name	Plan Name	Free Min	Night Min	Weekend Minutes	Price	Activation Fees	Night and Weekends	Contract Terms
AT&T	Cingular Nation [®] 4000 with Rollover [®]	4000	Unlimited	Unlimited	149.99	**** Activation Fees: \$36 Activation Fee for each new line, \$26 Activation Fee applies on each FamilyTalk line... full text	**** FamilyTalk line. Nights and Weekends: Nights are 9:00 p.m. to 6:00 a.m. Weekends are 9:00 p.m. Friday to 6:00 a.m. Monday full text	 All individual plans require a two-year contract.... full text
T-mobile	Get More Supr:	5000	Unlimited	Unlimited	129.99	**** Activation Fees: No Activation Fee for the new customers. full text	 Nights and Weekends: Nights are 9:00 p.m. to 6:00 a.m. Weekends are 9:00 p.m. Friday to 6:00 a.m. Monday full text	 anywhere across the US. All individual plans require a two-year contract.... full text
Our Product	National 4000	4000	Unlimited	Unlimited	199.99	**** Activation Fees: No Activation Fee for the new customers. full text	 contact required Nights and Weekends: Nights are 7:00 p.m. to 7:00 a.m. Weekends are 9:00 p.m. Friday to 6:00 a.m. Monday full text	 One year contact required Nights and Weekends: Weekends are 9:00 p.m. Friday to 6:00 a.m. Monday full text



Structured

Calling Plan	Minutes	Nights & Weekends	Weekly Price
Cingular Nation [®] 4000 with Rollover [®]	4000	Unlimited	\$149.99 per month
Cingular Nation [®] 5000 with Rollover [®]	5000	Unlimited	\$159.99 per month
Cingular Nation [®] 10000 with Rollover [®]	10000	Unlimited	\$279.99 per month
Cingular Nation [®] 20000 with Rollover [®]	20000	Unlimited	\$399.99 per month

Semi-structured

AT&T (Cingular) Website



T-Mobile Website

COMPETITIVE_ANALYSIS

COMPETITOR_PRODUCTS

ALL_PACKAGES

Internal Product Catalog

ATT_WIRELESS

T_MOBILE



Structured

Cingular Wireless Terms and Conditions

Cingular also imposes monthly a Regulatory Cost Recovery Charge, of up to \$1.25 to help defray costs incurred in complying with State and Federal telephone regulations; State and Federal Universal Service charges; and surcharges for customer-based and revenue-based state and local agreements on Cingular. These are not taxes or government-required charges.

Nationwide RAZR is \$199 (1000 Soft Card rebate) Offer: Limited-time offer. Purchase a Motorola RAZR or \$199 wireless handset for \$99.99 or more through a participating AT&T sales channel between 7/1/04 and 8/31/04 with a qualifying new 2-year Cingular Nation[®] plan. Web orders must be submitted by 8/29 to qualify. Only one Cingular Wireless 2-year Cingular Nation[®] 450 or greater rate plan, FamilyTalk[®] 700 or greater rate plan, or select promotional rate plan priced \$199.99 or greater per month that are weekend billed with eligible AT&T land service qualify. IT customers need

Unstructured



Enterprise Data Mashups

Demonstrating the Real Value of SOA



Data Services Reused in Many Applications: Customer Service Portal

R / servicios de cliente

JAVIER LOPEZ MATO [x] salir
español | galego

portada mis servicios usuarios facturas trámites soporte habla con R

BETA: envíanos tu opinión

servicios de cliente

mis servicios
gestión de los servicios que tienes actualmente contratados con **R**
[más información](#)

usuarios
modifica tus datos personales y gestiona nuevos usuarios asociados a tu contrato
[más información](#)

mis facturas
servicios de consulta de últimas facturas e información de consumo hasta la fecha
[más información](#)

trámites administrativos
una forma rápida de realizar muchas gestiones con **R** sin tener que llamarnos por teléfono
[más información](#)

soporte
ayuda, manuales, drivers y gestión de incidencias por avería
[más información](#)

habla con R
sugerencias, dudas, solicitud de información... ponte en contacto con nosotros para lo que necesites, usando un formulario o cualquiera de las otras formas de contacto disponibles
[más información](#)

gana un viaje a Londres

Applications: Customer Service Portal

[servicios contratados](#)
[descripción cuentas facturación](#)
[descripción de direcciones](#)

BETA: envíanos tu opinión

servicios contratados

titular del contrato con **R:** 76815731D - JAVIER LOPEZ MATO

organizar los productos y servicios contratados...

☒ por su dirección de instalación ☐ por la cuenta de facturación

C/ LAZARO CARDENAS 9 F 05 (15009) - CORUÑA (A) (A CORUÑA) ▼

[cambia descripción de esta dirección](#) | [cambiar todas](#)

1 producto(s) encontrado(s)

+ **combo3** [más info](#)

 **linea teléfono (981298493)** [más info](#)
servicios:

- identificación llamante [más info](#)
- contestador avanzado [más info](#)
- llamada en espera [más info](#)
- llamada a tres [más info](#)

 **internet** [más info](#)
servicios:

- velocidad 6 megas [más info](#)

Benefits and final considerations (I)

- **Reduction of response time** from the IT department in the **creation** of new **data services** to be used by the different business units (matter of hours instead of weeks/months) → **costs reduction**.
- **Exploit data** stored in any **internal** or **external** information system from a **single** platform (unique access **point** to all the information needed).
- Creation of added value applications that make **use** of **previous** information systems **infrastructure** without applying any modifications to it.

Benefits and final considerations (II)

- In the words of the Paco Fernandez of the Customer Service Portal:
 - “...Without the Data Mashups Platform we could not have tackled this project due to the great number of datasources we needed to access and how complex this task was...”.
 - “...**With Denodo Platform accessing datasources has become a trivial aspect to us**, now it's possible to access any of them through a unique interface. And moreover, in real-time! ...”.
 - “...On the other hand, because we are accessing critical data and systems from different departments, we needed to guarantee a controlled access to information. With Denodo Platform we access only the data we need, minimizing IT systems impact”.
- **DM Review:** Denodo Enables Spanish Cable Company to Gain Competitive Advantage Product Review published in DM Review Magazine July 2007:
 - “...**No other data integration product we looked at provided the breadth and flexibility of the Denodo Platform.** With the Denodo Platform, we are able to create a single view of customer data that encompasses data from any relevant sources. The product is **simple to use and quick to implement.** It has enabled us to create new business applications based on fresher, more complete data feeds...”
 - http://www.dmreview.com/article_sub.cfm?articleId=1087540



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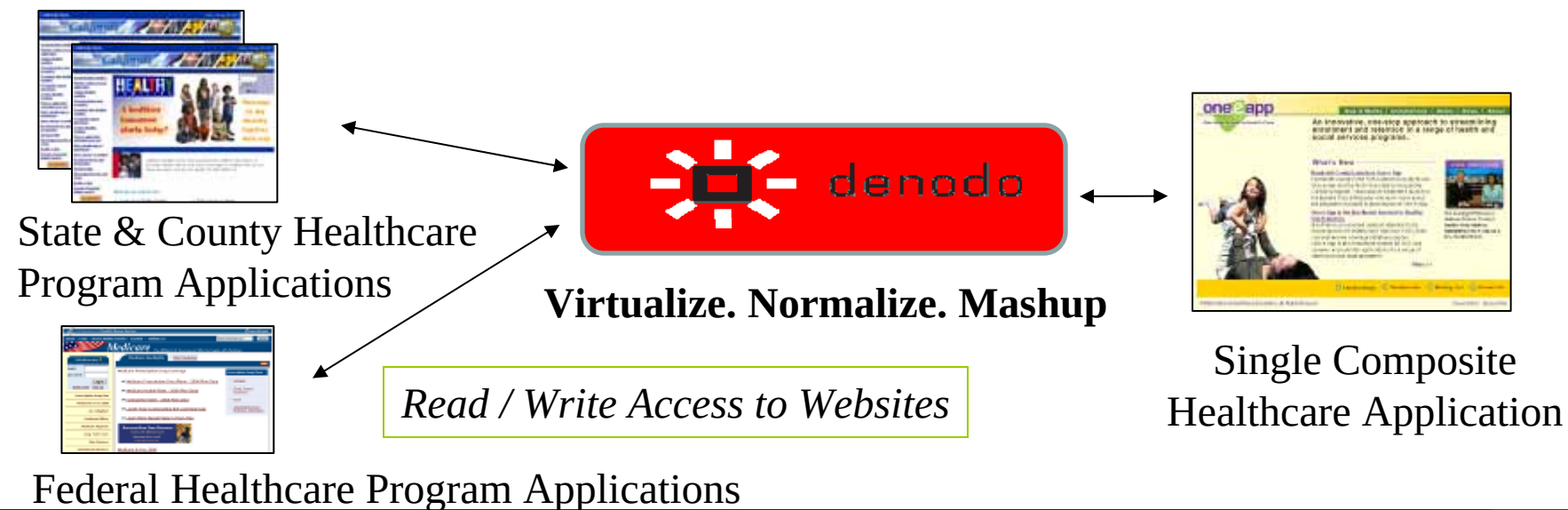
Evolution / Future Plans

- Integration on new departmental IT systems.
- Adding new features to:
 - Distributors Portal.
 - Integrations made on different departmental IT systems → providing detailed information about their customers and needs.
- Future evolution will be carried out by the client:
 - Graphical Interface / Application.
 - Automatic navigations.
 - Optimization for queries/views.
 - **Reduced TCO.**

Case Study: The Center to Promote Healthcare Access

– Make Data Services out of External / Partner Websites

- Visit www.oneeapp.org – One Stop Access to Healthcare Programs
- One central application for county-, state- and country-wide subsidized healthcare applications provides ease of use to agents and public
- Each individual program has its own application (usually web-based) with its own eligibility criteria – Denodo exposes them as bidirectional Services
- Win-Win: Unified access to agents and public users cuts costs while each healthcare system validates its own data, security and program rules.



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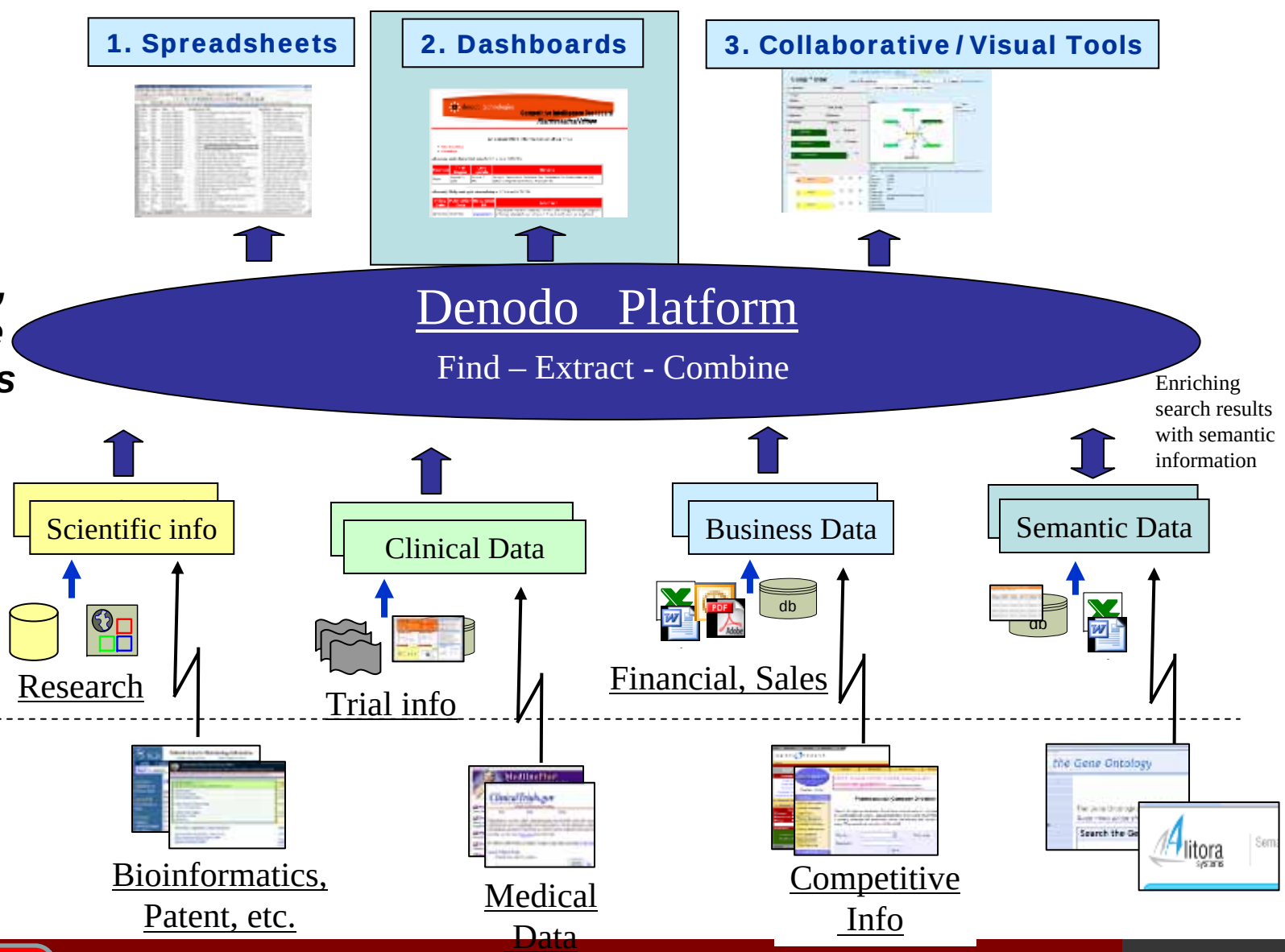
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Denodo's Pharma / Life Sciences Data Mashup Platform

*Used for BI,
Reporting,
Process
Enrichment,
Compliance
Applications*



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Which problems are Mashup Data Services best suited to solve?

- Numerous and diverse information sources (internal / external)
- Challenges around performance and availability for large applications
- Multiple sacrosanct systems of record (SOR)
- Information required in (near) real time, with high availability over a wide-area network (WAN)
- Integrated information requires model mapping and information quality scrubbing
- The resulting integrated view will serve many applications and usage scenarios
- Being able to search all kinds of data in your environment



Enterprise Data Mashups Technical Architecture



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Denodo Enterprise Data Mashups

Business Solutions

Data Virtualization
across all Sources

Enterprise Applications

BI, Portals, Dashboards

Web Applications

Denodo Platform

Right Information
at The Right Time

Denodo Platform

Publish: Virtual Real-Time Data Services

Mashup: Normalize, Transform, Relate

Connect: Read & Write Any Data Anywhere

Wrapper

Wrapper

Wrapper

Existing Data

Any Source
Any Format

Excel, XML, CSV
Database
Applications
Web Services

Hidden
Web



www

Web
Feeds

www



www

Web
Apps

PDF
Files
Emails
Documents

Structured Data

Web

Unstructured Info



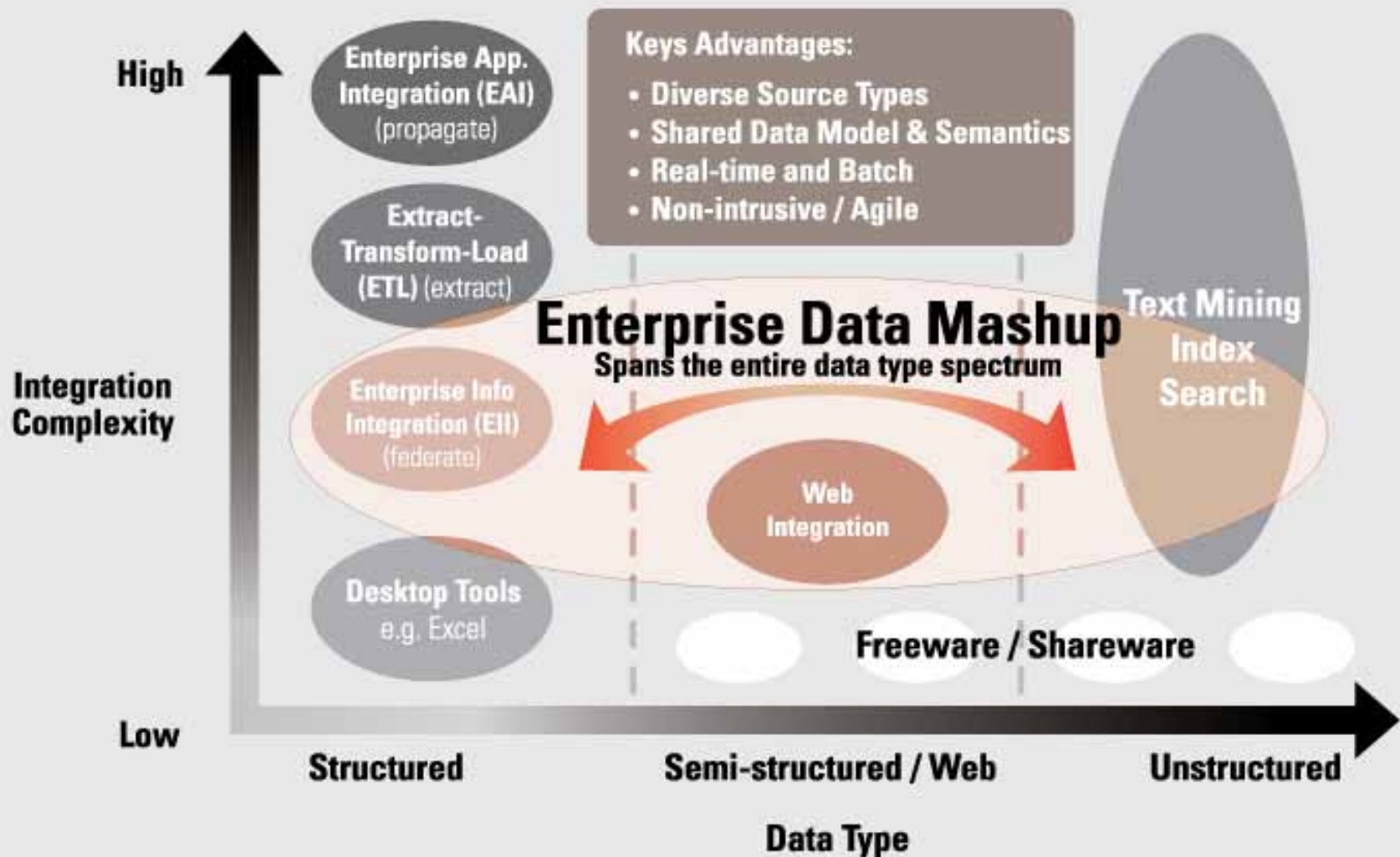
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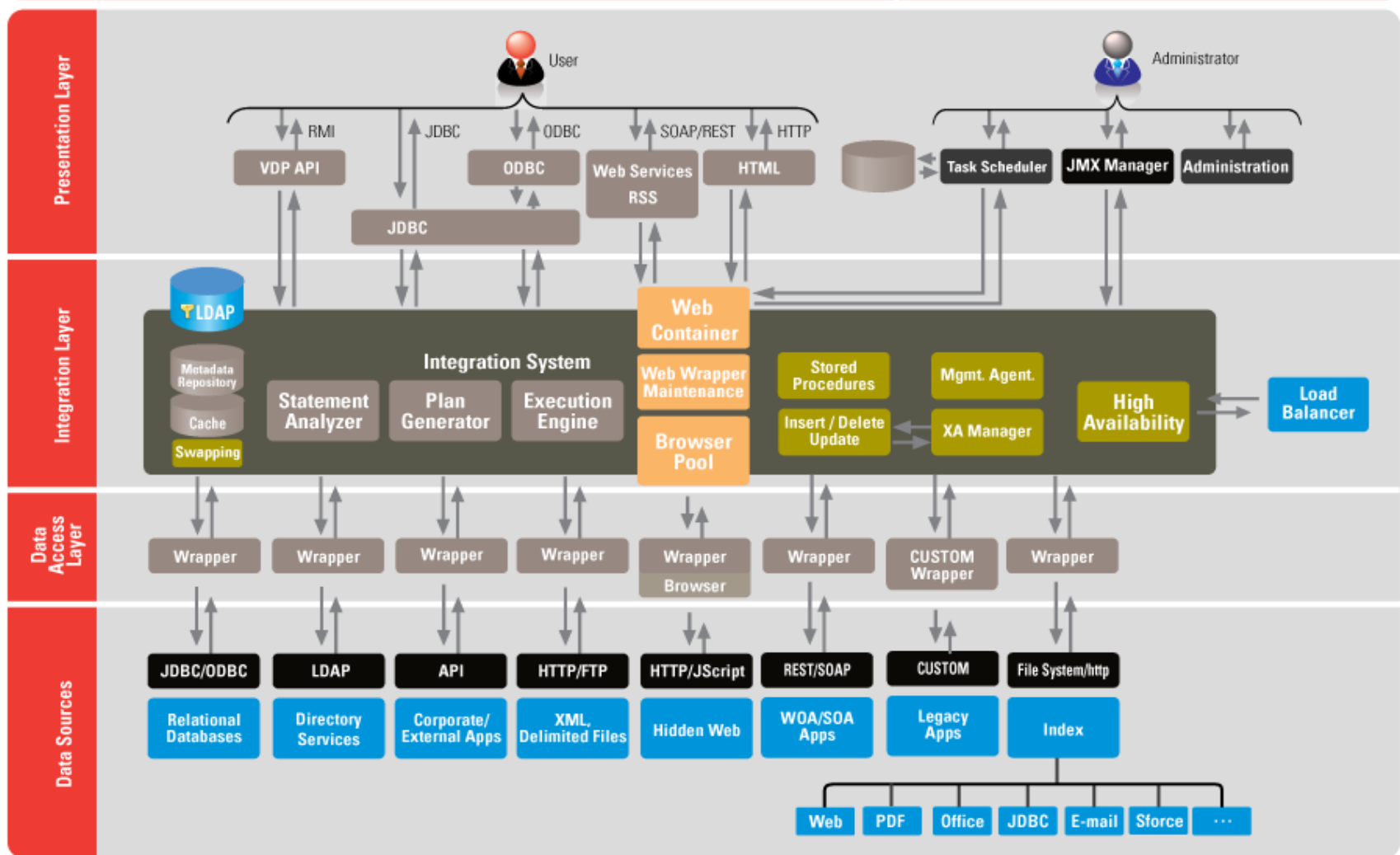


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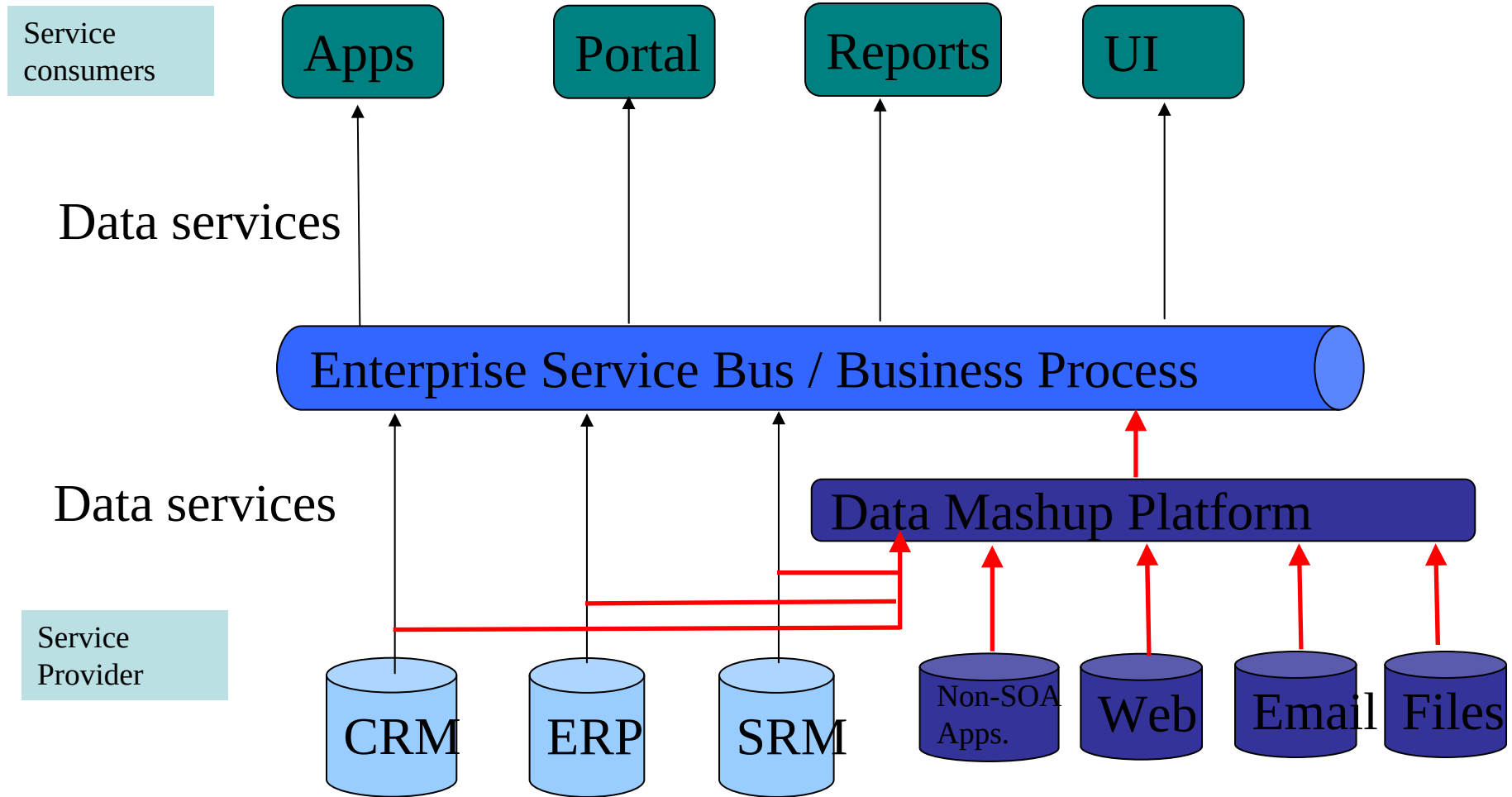
Denodo 'All in One' Single Platform



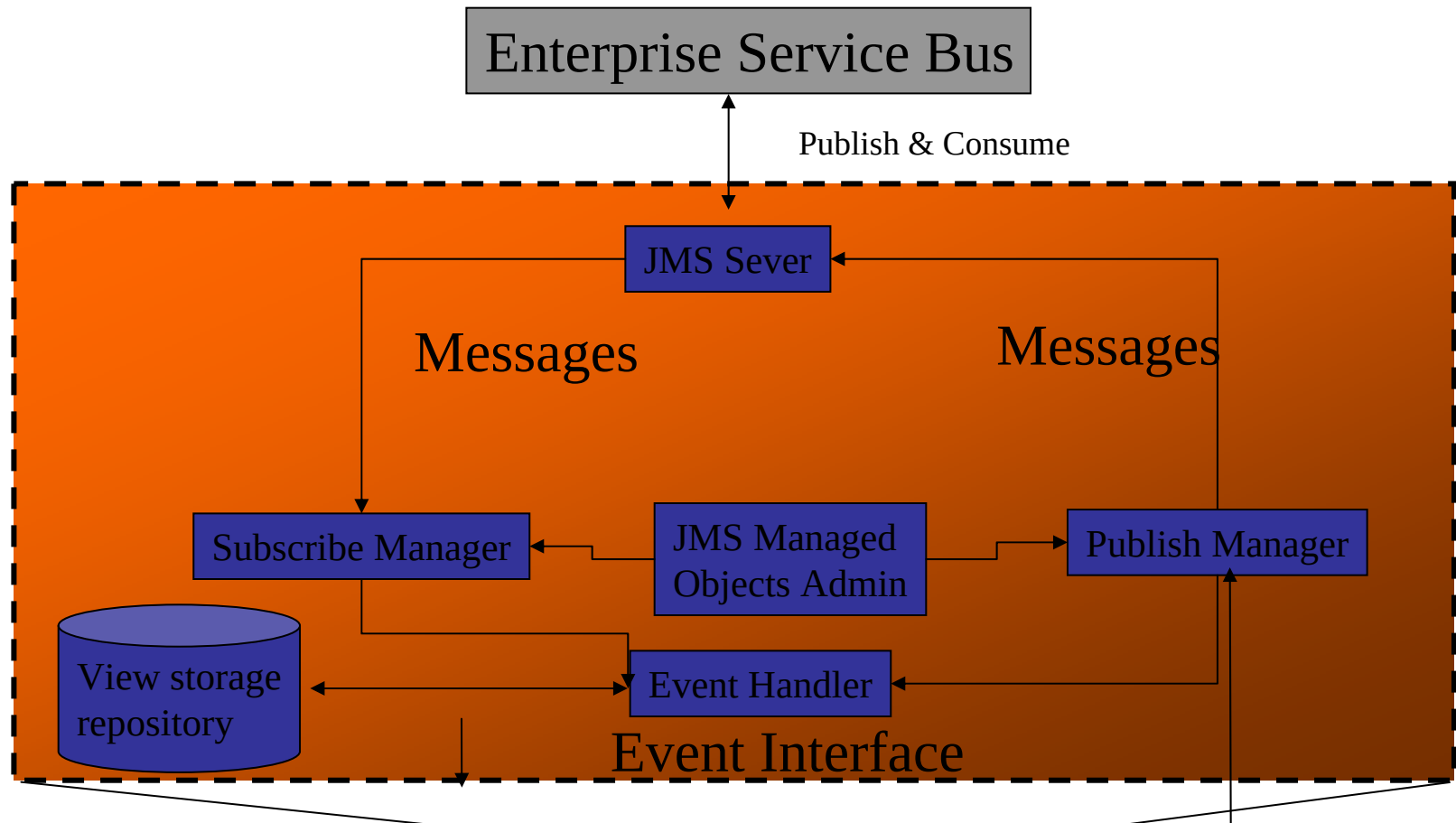
Denodo Detailed Architecture



Data Mashups are part of the Enterprise Ecosystem



Data Mashups interface with EDA JCA/JMS Connector Architecture



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Illustrating the Real Value of SOA



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Denodo in MDM Architecture - Data Federation for Golden Record

A. Access Web and Unstructured Data Sources and Normalize data for MDM to Process

B. Combine Structured data with Web and Unstructured data to create new data combinations to process



Build
Information
Relevance

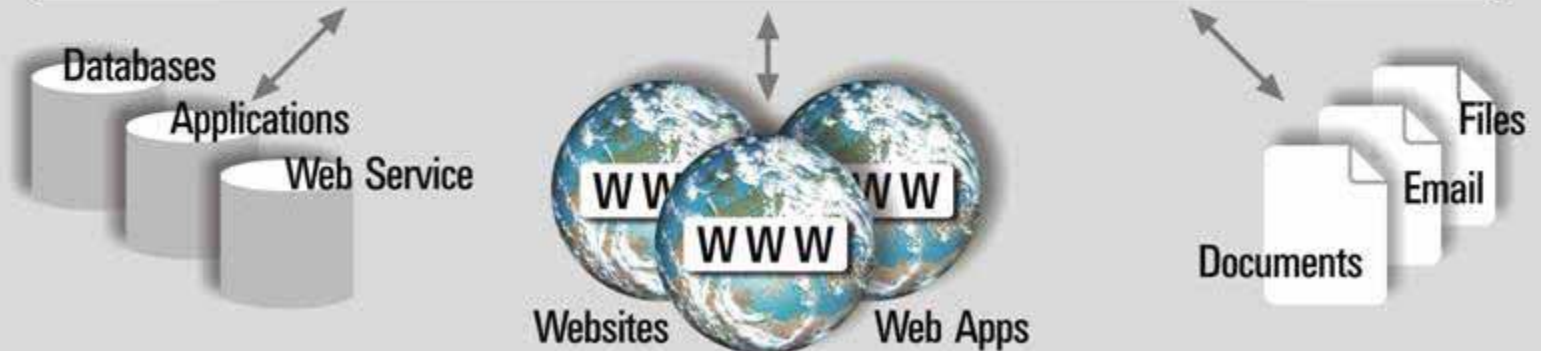
Denodo Platform

Multi-mode Access: Query, Browse & Search

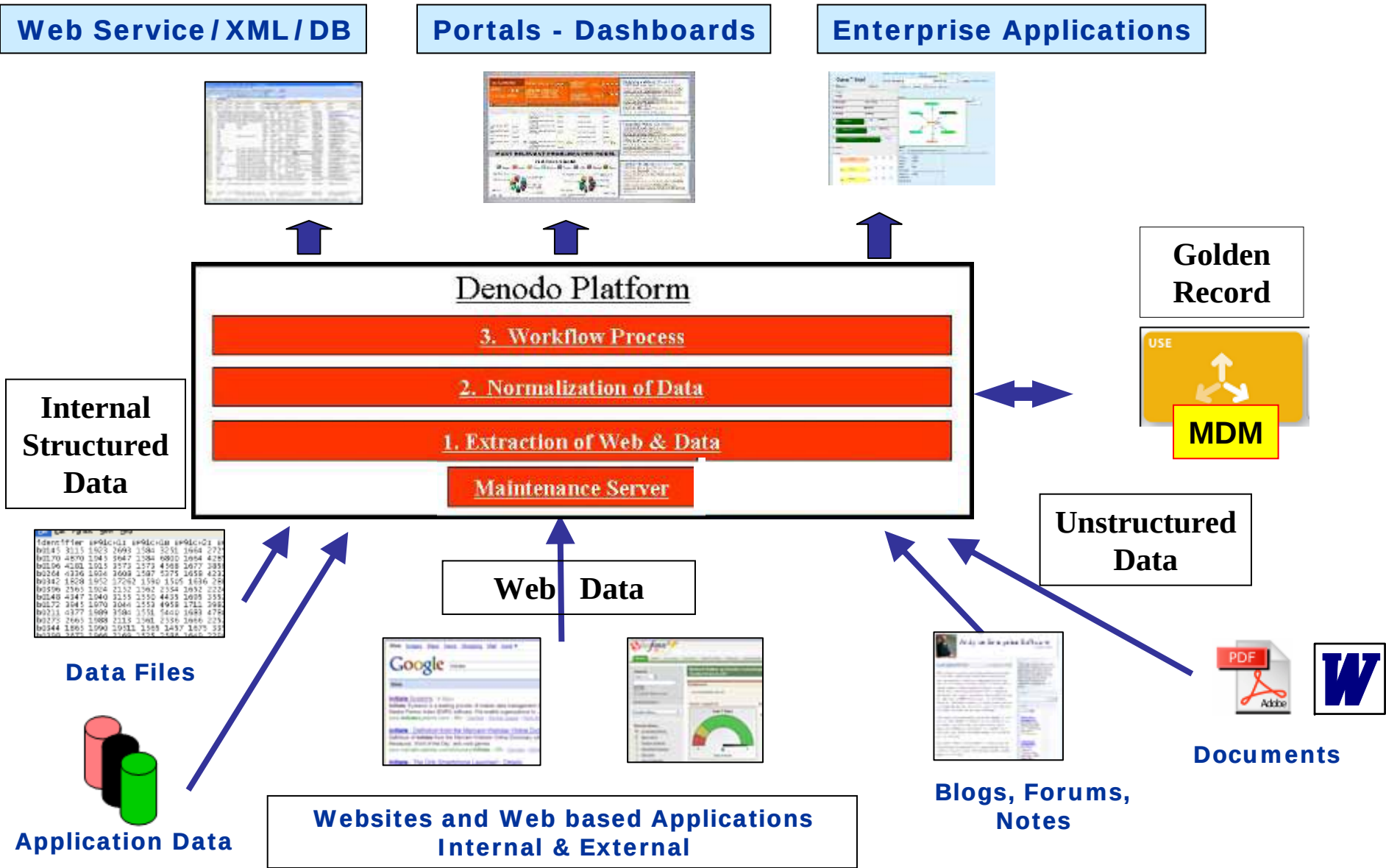
Process Engine: Transform & Relate Information

3 Data Engines: Read & Write

Access
Any Data
Source
Anywhere



Denodo in MDM Architecture - Single Entity View Enrichment



Enterprise Data Mashups

Conclusions & Next Steps



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Benefits from Enterprise Data Mashup

- Improve business responsiveness
- Increase information agility and visibility
- Reduce information infrastructure and integration efforts
- Expand SOA initiatives



Best Practices - From Denodo Customers

1. Mashups are for “end-users” and “systems” (SOA / Data Services)
2. Understand where the value is in your Mashups patterns
 - Visualization integration
 - Information integration
 - Process integration
3. Create a powerful Enterprise-class Mashup Infrastructure
 - Access to ALL information sources
 - Transform and Relate information as basic building blocks
 - Enable reusable Mashup Data Services
 - Add powerful Visual Mashup capabilities on top or feed existing portals
4. Project-based investment and ROI (while building infrastructure)



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Conclusion

- Alternative way of building and managing composite applications for faster projects
- Easy and fast to build, this expands the reach of data sources
- Beyond enterprise needs: customers, partners, ...

Thank You!

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- **Webinar on Data Mashups and SOA**
 - **Featured Speaker:** Mike Gilpin, VP Research Director, Forrester
 - **When:** Wednesday, December 17, 2008 10:00 AM - 11:00 AM PST
 - **To view webcast visit** www.denodo.com
- **Request a Custom Demo** of the Denodo Platform
 - Contact Denodo at 877-556-2531 or denodo.sales.us@denodo.com



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